

PROBLEM SET 1

ECON301—ECONOMETRICS I

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Instructions: Due on 1/2/2021; scan or type your work and submit it through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

1. The following exercise continues the example from Lecture 1. It will also help you remember some important prerequisite concepts from probability and statistics. Let X be a random variable that denotes the annual level of income, in thousands of dollars, from a small town. Let Y denote the annual level of savings, also in thousands of dollars. These random variables follow the following distributions:

$$X \sim \text{Gamma}(2, \lambda)$$

$$Y|X = x \sim \text{Uniform}[0, x]$$

where $\lambda = 0.7$. Recall that the Gamma distribution with parameters (α, β) is given by

$$f(x; \alpha, \beta) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)} 1_{[0, \infty)}(x)$$

where $\Gamma(\cdot)$ is the Gamma function. The notation $Y|X = x$ is referring to the conditional random variable with distribution $f(y|x)$ given by the uniform distribution over $[0, x]$, i.e.,

$$f(y|x) = \frac{1}{x} 1_{[0, x]}(y)$$

Note, as in the example from the lecture, that by construction we have that $Y \leq X$.

- Using the law of total probability, compute the joint distribution of (X, Y)
 - Compute the marginal distribution $f_Y(y)$.
 - Compute the marginal distribution $f_X(x)$.
 - Compute the conditional expectation $E[Y|X = x]$.
 - Compute the conditional expectation of Y when $X = 2.5, 5, 10$. Compare with the simulation results from lecture 1.
 - Verify the law of iterated expectations for this example by:
 - Using the marginal distribution of Y , $f_Y(y)$, compute $E[Y]$.
 - Looking at $E[Y|X]$ as a random variable of X , and using the marginal distribution of X , $f_X(x)$, compute $E[E[Y|X]]$.
 - Compute $\text{Var}(Y|X = x)$.
 - Verify the law of iterated variance for this example by:
 - Using the marginal distribution of Y , compute $\text{Var}(Y)$.
 - Using the marginal distribution of X , compute $E[\text{Var}(Y|X)]$ and $\text{Var}(E[Y|X])$.
 - Compute the best linear predictor $E^*(Y|X) = \alpha + \beta X$ by using the formulas in Thm. 2.7.
 - Verify the results from Thm. 2.9 by using items (d) and (h). Conclude that since the conditional expectation is linear, it is also the best linear predictor.
2. Prove Thm 2.4 from Lecture 2. *Hint:* $\text{Cov}(X + Y, X + Y|Z)$.
3. Let (X, Y) be a r.v. with pdf $f(x, y) = (x + y)1_{[0, 1]}(x)1_{[0, 1]}(y)$, show that
- $$E(X^m Y^n) = \frac{(m+1)(n+2) + (m+2)(n+1)}{(m+1)(m+2)(n+1)(n+2)}$$
4. Let $U^* = Y - E^*(Y|X_1, \dots, X_k)$, show that
- $$E(U^*|X_1, \dots, X_k) = E(Y|X_1, \dots, X_k) - E^*(Y|X_1, \dots, X_k),$$
- and provide an example where the above expectation is different from zero.
5. Let (X, Y) be a random vector with pdf $f(x, y)$. Let $Z = E[Y|X]$. Show that the BLP of Z as a function of X is the same as the BLP of Y as a function of X . Interpret intuitively this result.